

Maksim Sviridov

sviridov@umich.edu | (248) 882-5027
([LinkedIn](#) | [Portfolio Website](#) | [GitHub](#))

Education

University of Michigan Ann Arbor

Masters of Science in Engineering - Computer Science

August 2025 - May 2026

- **GPA:** 3.72
- **Coursework:** Advanced Computer Vision, Advanced Database Systems, Sensors and Automation

Bachelor of Science in Engineering in Computer Science

Sept. 2022 - May 2025

- **GPA:** 3.65
- **Coursework:** Distributed Systems, Operating Systems, Machine Learning, Formal Verification, Search Engines

Skills

- **Programming Languages:** C++, Python, CUDA, Golang
- **Frameworks/Libraries:** PyTorch, HuggingFace, ROS2, OpenCV, OpenAI Gym, MuJoCo, Gazebo
- **Tools:** Docker, Git, Bash, Google Cloud Platform, Linux, CMake
- **Languages:** German, Russian

Work Experience

DELMIA R&D, Dassault Systèmes

Robotics Software Engineer Intern

Royal Oak, Michigan

June 2026 - Present

- Developed a GPU-accelerated LiDAR simulation framework in C++/CUDA using Nvidia cuBQL, achieving a 113× speedup over an optimized CPU baseline (tinyBVH) across scenes up to 2M triangles at 512K rays/query
- Built an autonomous weld-seam trajectory correction pipeline that identifies parts via OCR, locates and extracts the weld seam using edge detection on RGB-D scans, then fuses the measured seam with the nominal digital-twin trajectory to generate a corrected path accounting for real-world part and fixturing error
- Created a Python inference server for SAM-6D pose estimation, with a CAD registration API auto-generating reference views via BlenderProc and a backend supporting multiple segmentation and pose estimation models
- Bridged a standalone C++ application to ROS2 over FastDDS middleware by generating interface definitions and their C++ codegen structs to publish/subscribe to ROS2 topics from outside the ROS2 process space

DELMIA R&D, Dassault Systèmes

Software Engineer Intern

Royal Oak, Michigan

May 2025 - August 2025

- Built a synthetic data generation pipeline in Python to programmatically augment large teleoperation datasets with configurable texture, lighting, material, and trajectory transformations to improve dataset robustness
- Developed infrastructure to benchmark Vision-Language-Action (VLA) models for industrial manipulation tasks across latency, accuracy, and deployment constraints
- Designed real-time perception pipelines in ROS2 for industrial object detection, pose estimation, and motion planning and validated them in Mujoco simulations

Project Experience

C++ Distributed Search Engine ([Github](#))

- Built a distributed search engine in C++20 from scratch including a crawler, HTML parser, URL frontier, inverted index, query parser, and ML ranking engine that indexed 50M+ web pages with subsecond query latency on GCP
- Designed a custom binary-encoded inverted index to store and query 100GB of documents with minimal memory overhead and fast on-disk lookup
- Architected sharded URL crawling to parallelize ingestion across nodes while minimizing inter-node data egress, enabling cost-efficient crawling at scale

Symposiums

Learning Composable NBA Player Embeddings from Possession-Level Data ([PDF](#))

[MSSISS 2026](#)

Ann Arbor, Michigan

March 2026

- Presented a novel Transformer architecture at MSSISS 2026 trained on over 25 million possession-level events to learn continuous player representations, enabling downstream inference on arbitrary player and team statistics
- Designed masked token prediction and cross-attention objectives to isolate individual player impact from their team
- Addressed cold-start learning by incorporating physical player attributes into the latent embedding space